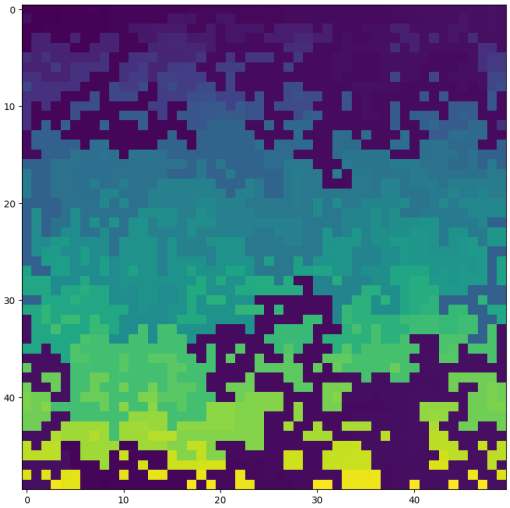


Final Report

The Art of Scientific Computing: Deffuant Model of Opinion Dynamics

Author:
Nathan Rees

Subject Handbook code:
COMP-90072



Faculty of Science
The University of Melbourne

Original Version: 22th October 2026

Subject co-ordinators: A/Prof. Suzie Sheehy & Prof. Christian Reichardt

Contents

1 Introduction **2**

2 Implementation **3**
2.1 Lattice Generation 3
2.2 Deffuant Model 4
2.3 Measurements on the System 5

3 Discussion **7**
3.1 Measurement Outcomes 7
3.2 Cluster Behavior 10
3.3 Phase Transition 13
3.4 Conclusion 21

Appendix A Extensions to Deffuant Model **22**

Bibliography **22**

Chapter 1

Introduction

Physical models of systems such as the Ising model of ferromagnetics and entropic models for system evolutions have long existed inside the realm of statistical physics. These models have been highly successful in describing macroscopic phenomena based on key assumptions and simplifications about the microscopic nature of the physical world. Throughout the past century, many of these principles have been applied to non-physical systems. Great success has been found especially in describing economics, even being given the name "econophysics" [4]. More recently, these principles have also been applied to the study of how opinions disseminate throughout a population. This field has adopted the name of "opinion dynamics", which aim to describe how opinions (whether discrete or continuous) change throughout a population depending on how agents communicate with one another. While this report focuses on a system that may most resemble how a continuous political ideology changes throughout the system, these principles from opinion dynamics are also applicable to other social dynamics such as language and cultural dynamics.

In opinion dynamics there are many different models that one can consider. One of the simplest is the voter model [3]. Agents in a lattice (or network for a slightly more complicated system) take one of a discrete value (typically one of two values), and agents either change or don't change their value based on a rule (typically dependent on the values of the neighbours). For example, an agent may change their value if two or more of its neighbours have a different value. While this model is fairly simple to implement, it makes many approximations and assumptions that fail to reflect its real-world inspiration. Values of agents are typically more nuanced than a simple discrete value (such as a simple "yes" or "no" to an opinionated question). Instead, one can allow for a continuity of opinion (this can be represented as a bounded interval, such as from 0 to 1 or from -1 to +1) to encapsulate agents' opinions not being binary, but only allow for agents to communicate to their neighbours a binary opinion. This is known as a CODA (continuous opinion, discrete action) model. While this better reflects its real-world inspiration, it still fails to encapsulate how agents may strongly or weakly communicate agreement or disagreement, and also how agents may not communicate at all if difference in opinion is too great.

The Deffuant model (sometimes also referred to as the Deffuant-Weisbuch model) [5] aims to implement these into opinion dynamics. The Deffuant model is known as a bounded confidence model [1] Typically, analysis for this model is conducted in a network but for this report a two dimensional lattice will be used. The Deffuant model encodes agents' opinions into a bounded interval (for this report we will take this bound as being from 0 to 1 inclusive). Given a random agent and a random neighbour of the agent, the two communicate via the rule

$$x_i(t+1) = x_i(t) + \mu[x_j(t) - x_i(t)] \tag{1.1}$$
$$x_j(t+1) = x_j(t) + \mu[x_i(t) - x_j(t)] \tag{1.2}$$

where x_i is a random agent, x_j is a random neighbour of the agent and μ is called the convergence parameter, and is based on a compromise strategy [1]. There is also an added requirement that

$$|x_i(t) - x_j(t)| \leq r \tag{1.3}$$

where r is a threshold whereby agents won't communicate unless the above is satisfied and will be referred to as the range [5].

Chapter 2

Implementation

2.1 Lattice Generation

While typically one could simply generate a lattice or network at random without any weighting, it is far more interesting to be able to have different starting conditions whereby agents are more likely to be extreme, centralist or any other distribution one wants to test. As such, one can specify their own lattice for analysis or use the beta distribution with varying α and β parameters to produce different initial lattices. Interesting beta distributions to try include the uniform distribution ($\alpha = 1$, $\beta = 1$), a more polarised distribution (that is a distribution with higher probability density at the extremes close to 0 and 1) ($\alpha = 0.5$, $\beta = 0.5$) and a more centralist distribution ($\alpha = 2$, $\beta = 2$).

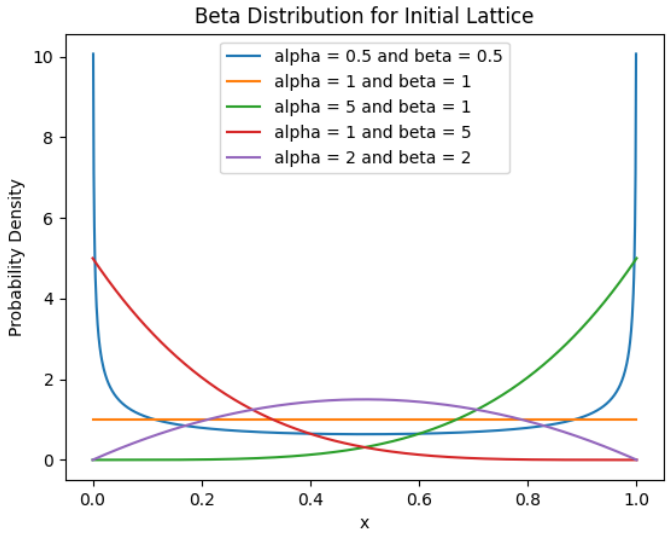


Figure 2.1: The probability density function for various values of α and β used in generating the initial lattice. This shows how the initial lattice skews, whether its central or polarised.

The code utilises the numpy.random.beta function to produce a lattice of given dimensions (default 50 by 50 if no lattice is specified). An example inital voter lattice is shown below.

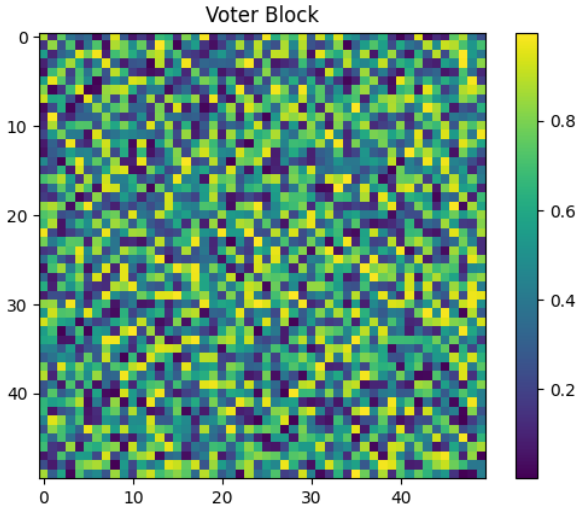


Figure 2.2: Example of a Voter Block with $\alpha = 1$ and $\beta = 1$

The lattice generates values from 0 to 1 for each site based on the beta distribution specified. Figure (2.2) uses $\alpha = 1$ and $\beta = 1$, so every value is equally likely due to the beta distribution being a uniform distribution for these values. An example of a lattice generated with different α and β is shown below.

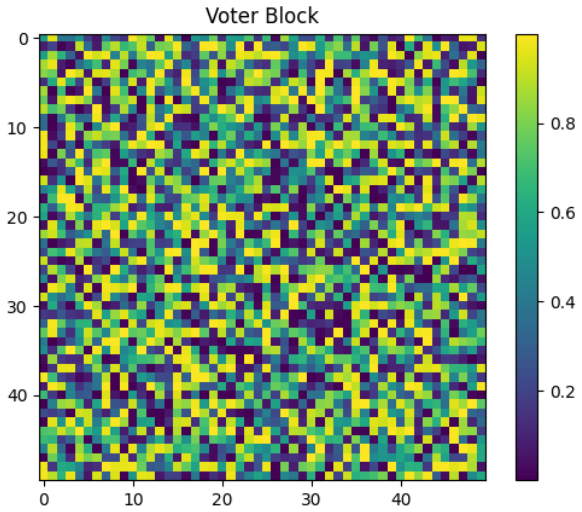


Figure 2.3: Example of a Voter Block with $\alpha = 0.5$ and $\beta = 0.5$

Note how values closer to 0 and 1 are more common for a lattice where the beta distribution used to generate it has $\alpha = 0.5$ and $\beta = 0.5$ than for the lattice where $\alpha = 1$ and $\beta = 1$.

2.2 Deffuant Model

Two different Deffuant models were developed for the code used in this report. One model (which will be referred to as the Deffuant model or normal Deffuant model) simply picks two agents at random (they do not need to be neighbours) and applies the rules/equations 1.1,1.2 and 1.3.

The second developed model is different in how it chooses the two agents. An agent is chosen at random, but one of its four neighbours are chosen at random (without bias) instead of a completely unrelated agent. Then the above rules are applied for updating the lattice. This will be referred to as the close Deffuant model.

A variation of the Deffuant model (which will be dubbed the noisy Deffuant model) was also developed. Instead of the threshold being a specified value, it is instead chosen from a beta distribution at random with

a variable change so that a mean and variance can be specified instead of a value for α and β. First, α and β were re-parametrised by the equations

$$\alpha = \phi\mu \quad and \quad \beta = \phi(1-\mu) \tag{2.1}$$

where μ is the mean and ϕ is defined as the precision [7]. This is further re-expressed in terms of a mean and a standard deviation as to resemble a more typical normal distribution via

$$\phi = \mu\frac{1-\mu}{\sigma^2} - 1 \tag{2.2}$$

Depending on the specified model, the lattice is iterated many times to develop a new lattice. The number of iterations (also referred to as time steps) can be specified by the user, or instead a tolerance can be specified such that if the difference in the variance of the lattice between the latest time step and the time step that happened 4 time steps ago is less than the tolerance, the loop stops. Calculation of values such as variance is discussed later.

All models assume periodic boundaries. This is so that the behaviour of sites is the same throughout the lattice, with no changes in behaviour at the edges or corners of the lattice. Discrete voter models were also toyed with initially but these were more to test the methodology for the implementation of the Deffuant model and as such won't be discussed.

2.3 Measurements on the System

There are a number of metrics that were developed to measure how the lattice evolves and to compare the three different model variations. Comparison to neighbouring sites, variations in the values sites take and how clustering occurs in the lattice after sufficient time were all considered important for the analysis of the three models, and as such multiple measures (discussed below) were made to try and quantify this behaviour.

2.3.1 Average Difference To Neighbours

The difference to neighbours (DN) for an agent at lattice site [i,j] is defined to be

$$DN[i,j] = |\frac{x[i-1,j] + x[i+1,j] + x[i,j-1] + x[i,j+1]}{4} - x[i,j]| \tag{2.3}$$

In words, the DN for an agent is the absolute difference between the average of the agent's neighbours and the agent. This is then averaged over all agents to find the average DN (ADN).

2.3.2 Variance

Variance has the typical meaning in this report as in statistics [6]. The lattice is squared element-by-element and then averaged using numpy, and this is subtracted from the mean of the lattice squared, as per the equation

$$\sigma^2 = E[X^2] - E[X]^2 \tag{2.4}$$

This is typically taken at each timestep, and is a good measure for how much the system is changing over time and hence is used to limit the number of iterations required for updating the lattice. This is done to reduce compute time.

2.3.3 Cluster Analysis

Some configurations of parameters result in clusters of values, and as such measurements of these clusters would be insightful. Variance and ADN may capture some of this information, but for better analysis the clusters need specific measurement and analysis tools. To do this, the lattice first has a copy made where agents now take on new values depending on the value of r (or the mean for the noisy Deffuant model). It was observed that the number of values the clusters (which will be referred to as NVC) evolved towards roughly followed the below (for $\alpha = 0.5$, $\beta = 0.5$)

$$NVC \approx round(\frac{1}{r}) - 1 \tag{2.5}$$

where round is the rounding function that rounds to the nearest integer. From this, the range of values the agents can take is split evenly and agents falling into a range is given the corresponding value. For example, if r = 0.25, then $NVC = 3$ and so if an agent has value 0.85, the copy of the lattice has the agent at the same position assigned a value of 3 as it lies in range $\frac{2}{3}$ to 1. If the agent were to have value 0.51 the copy agent

would be assigned value of 2 as the agent has value that lies in between $\frac{1}{3}$ and $\frac{2}{3}$. If the agent were to have value 0.16, the copy agent would have value 1 as the agent has value that lies between 0 and $\frac{1}{3}$.

This is only applicable for r values less than 0.5 however as this starts to estimate only one or even zero clusters beyond this. As such a manual threshold was implemented such that values between 0 and 0.2 were considered in the same group, values between 0.2 and 0.4 as part of the same group, 0.4 to 0.6 as part of the same group, 0.6 to 0.8 part of the same group and 0.8 to 1 as part of the same group so that there are 5 total groups. It is worth noting that literature states that a phase transition (or a critical point) occurs at $r = 0.5$ [8] where if $r > 0.5$ the network or lattice is expected to converge to consensus (that is network/lattice agents take practically identical values across the whole network/lattice), hence why this manual defining of group clusters is necessary beyond $r > 0.5$.

Chapter 3

Discussion

3.1 Measurement Outcomes

The μ term in the communication rule ultimately had little effect on measurement outcomes in the long term. The main effect of the μ term is that it makes the system reach its steady state at a different time scales. Typically, a μ value of around 0.5 produces the steady state quickest. Extreme values for μ simply make the update rule do nothing (for $\mu = 0$) or switch values of sites (for $\mu = 1$), and as such these values are not considered.

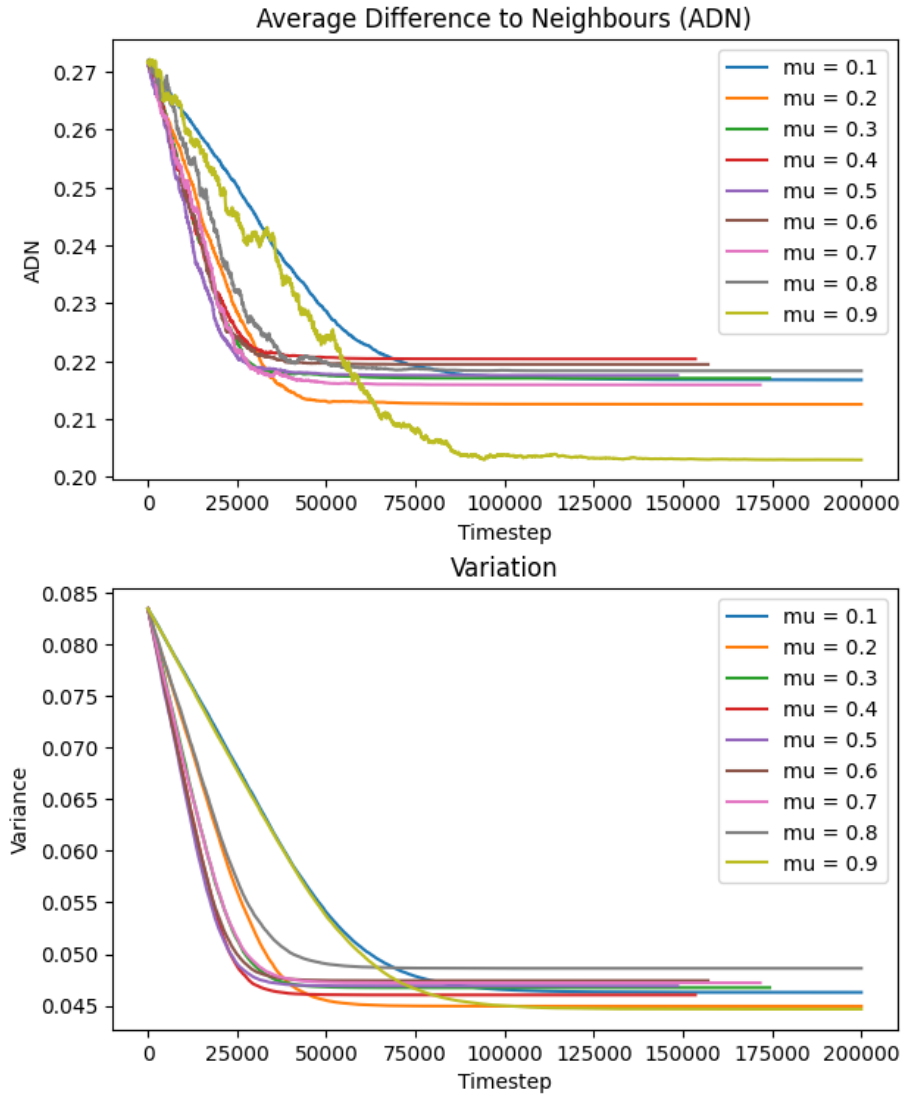


Figure 3.1: The average difference to neighbours (ADN) and variance as the lattice is iterated on. As this is a random process there will be some variation in the ADN and variance between μ values. μ values near 0.5 tend to reach equilibrium quicker than μ values near 0 and 1

A μ of 0.4 is used for discussion of other behaviors such as how the r value changes the system. This is done to set a standard between different lattices and to also ensure less iterations are necessary to achieve steady state. This behavior for the μ is also reflected in the literature, it was expected to have little impact on long term iterations of the lattice and the findings of this report support this idea.

Another feature that doesn't affect the measurement outcomes is the dimensions of the lattice (assuming extremely small dimensions such as 1x1 or 2x2 lattices are not used). Assuming sufficiently large lattices, the amount of time steps to reach steady state increases, but the behavior at steady state does not change. This mimics how μ doesn't necessarily change the long term behavior, simply how long it takes to see this long term behavior. This is both for square lattices (which will be used as a baseline throughout the discussion) and for rectangular lattices. Clearly a differently-sized lattice will affect the area of clusters and the total number of clusters as there are more sites, but the behaviour of the phase transition (which will be discussed later) is almost identical as shown below. It appears that the area and number of clusters scales linearly with the size (or area specifically) of the lattice, which is unsurprising considering how this system is supposed to be scalable. Variance and ADN appear almost identical as these are unaffected by areas, only local behaviours within the lattice.

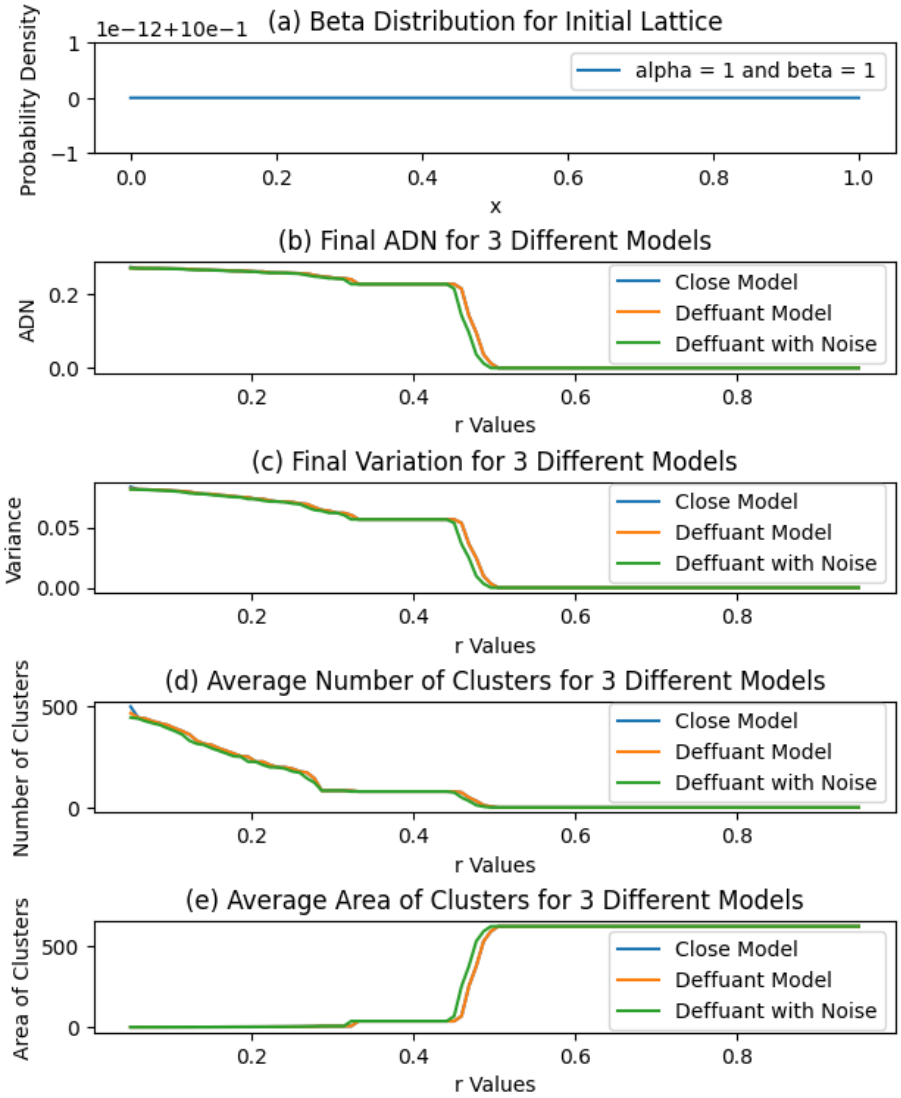


Figure 3.2: Various measurements on a 25 by 25 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 1$ and $\beta = 1$ results in a uniform distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

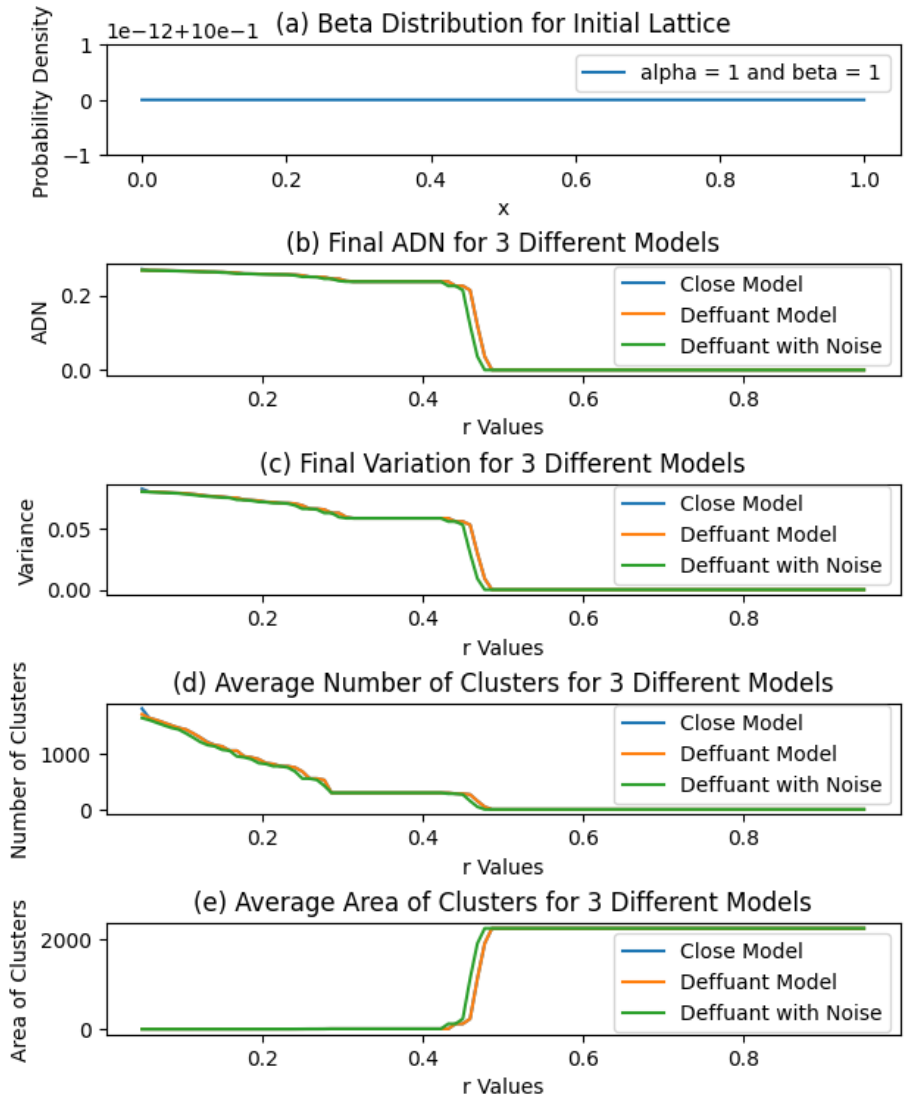


Figure 3.3: Various measurements on a 30 by 75 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 1$ and $\beta = 1$ results in a uniform distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

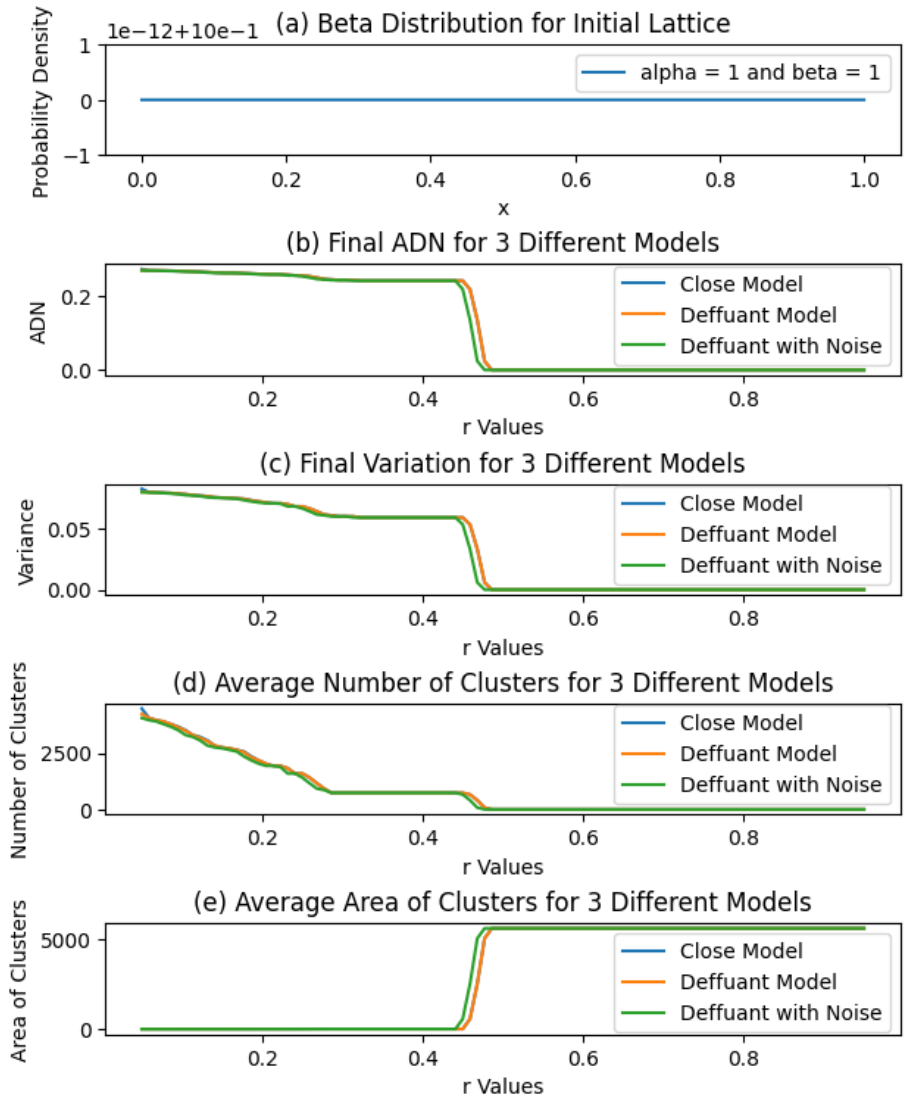


Figure 3.4: Various measurements on a 75 by 75 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 1$ and $\beta = 1$ results in a uniform distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

It is clear however from the above that there is a difference between how the noisy Deffuant model behaves compared to the close Deffuant model and the normal Deffuant model. As this mostly pertains to the phase transition however it will not be discussed here.

3.2 Cluster Behavior

There are two regimes to consider, one regime where the r value for the update rule is below the phase transition and one where the r value is above. The phase transition will be discussed later.

3.2.1 r Values Below Phase Transition

For lower r values below the phase transition r value, the lattice sites will tend to a finite number of values based on the r value specified. This was previously given the name NVC.

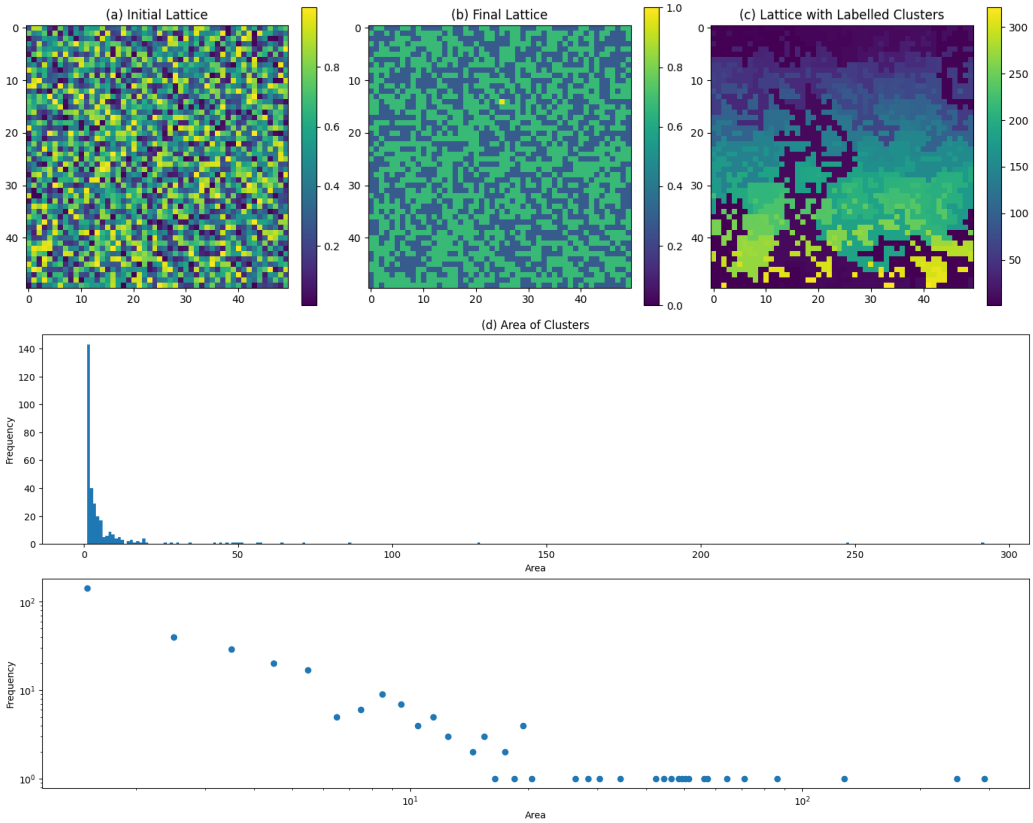


Figure 3.5: (a) The initial lattice was generated with $\alpha = 1$ and $\beta = 1$. (b) The lattice was iterated through using the Deffuant model with $\mu = 0.4$ and $r = 0.25$. (c) The lattice with site values being the label of the cluster the site is a part of. (d) The area of the cluster(s) as both a histogram and a log-log scatterplot.

It is worth noting that while the figure above may look like it only has two major values that the lattice sites take, there are other ones that are around 0.5, hence the use of NVC to try and capture these sites. Typically below the phase transition, all models follow NCV approximately well and as such one can approximate the long term behaviour for r values lower than the phase transition threshold. This is only approximate however and based on observation, it is not as certain as the behaviour above the phase transition threshold. Regardless, for lower r values the lattice always exhibits this splitting of lattice site values, regardless of μ values (assuming μ is not an extreme value like 0 or 1) and assuming the lattice generated uses a symmetric beta distribution (such as $\alpha = 0.5$ and $\beta = 0.5$).

For a lattice that is generated from an asymmetric beta distribution, the lattice reaches consensus with dots of lattice sites that cannot reach consensus.

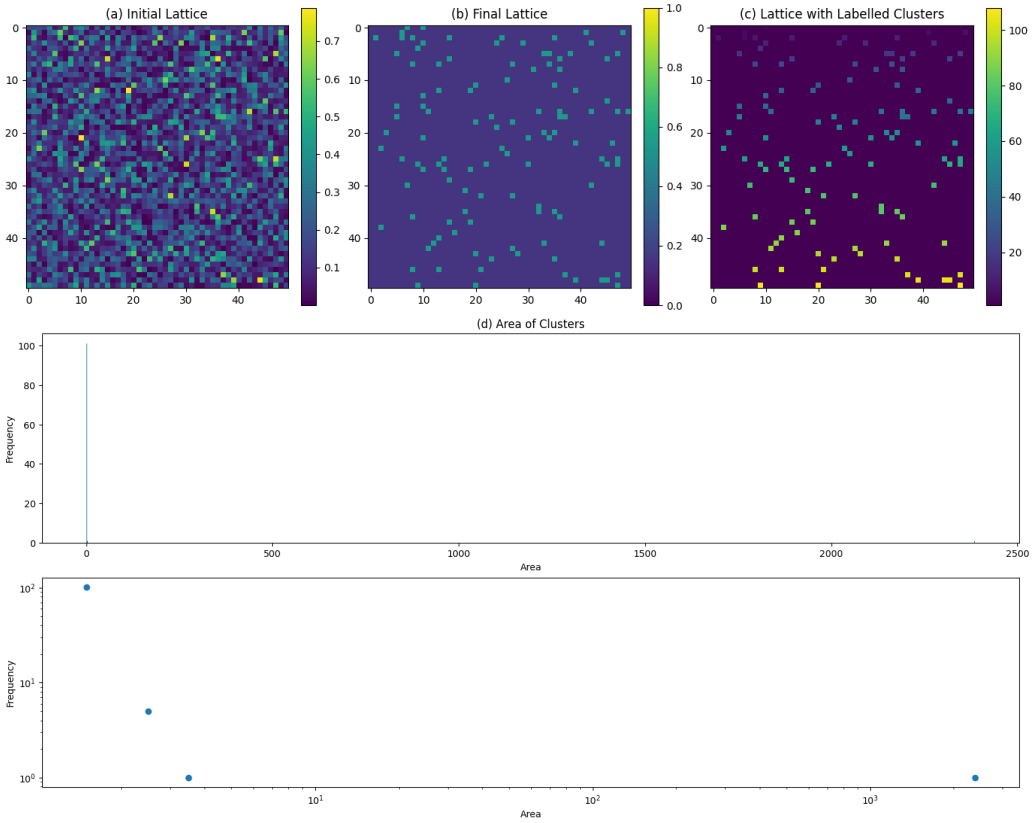


Figure 3.6: (a) The initial lattice was generated with $\alpha = 1$ and $\beta = 5$. (b) The lattice was iterated through using the Deffuant model with $\mu = 0.4$ and $r = 0.25$. (c) The lattice with site values being the label of the cluster the site is a part of. (d) The area of the cluster(s) as both a histogram and a log-log scatterplot.

Perhaps unsurprisingly, if the lattice is already dominated by one side of the bound, that domination will be reflected as the lattice is iterated on. Since there will be some values generated outside of the dominating values (at a much lower rate), those lattice site values will not be able to communicate with the majority, causing spots that do not agree with the majority of the lattice, hence the spotty iterated lattice.

The number of clusters trends downwards as r increases towards the phase transition. As r increases, more lattice sites are able to communicate with each other, resulting in the values the lattice sites can take decreasing and therefore the number of clusters also trends downwards.

3.2.2 r Values Above Phase Transition

For higher r values above the phase transition, the lattice tends to reach consensus. This occurs for the close Deffuant model, Deffuant model and noisy Deffuant model, assuming that the r value is larger than the phase transition r value.

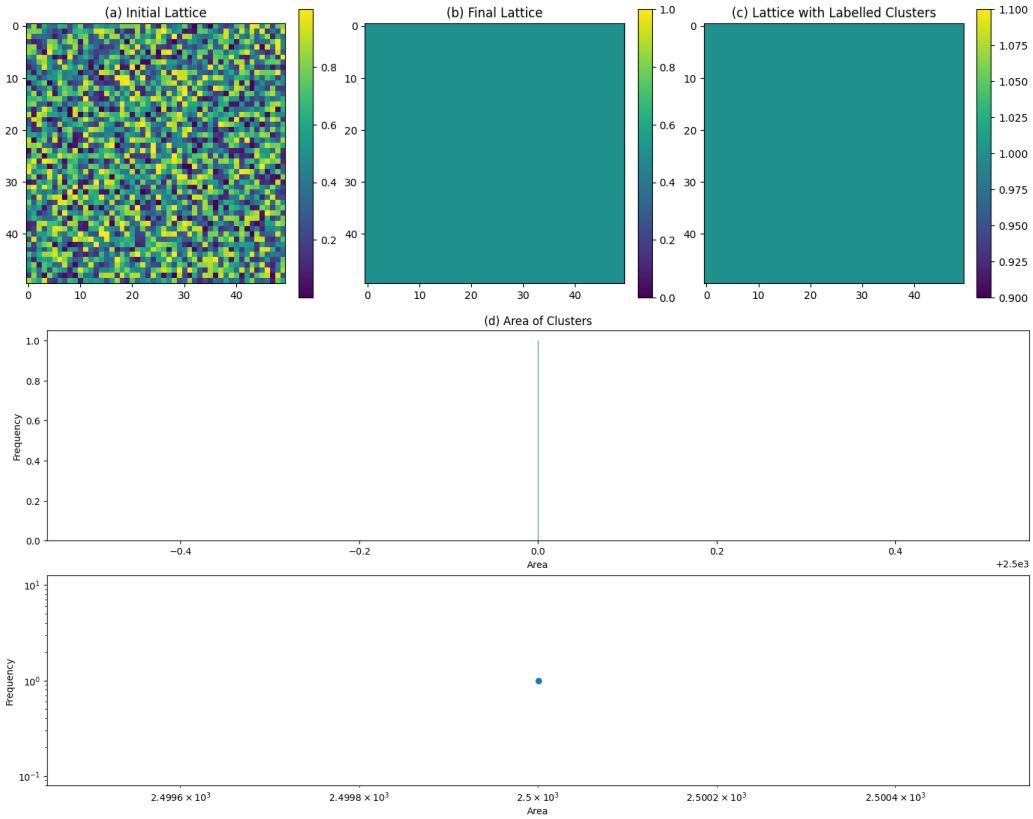


Figure 3.7: (a) The initial lattice was generated with $\alpha = 1$ and $\beta = 1$. (b) The lattice was iterated through using the Deffuant model with $\mu = 0.4$ and $r = 0.8$. (c) The lattice with site values being the label of the cluster the site is a part of. (d) The area of the cluster(s) as both a histogram and a log-log scatterplot.

From the above figure, one can see that it approaches consensus with all sites being extremely close in value to each other to the point of forming on large "cluster" (although the word "cluster" loses meaning in this case as its just one value for the whole lattice). This happens practically every time for large enough r value and as such the exact long term behaviour is a lot more predictable than that for low r values.

3.3 Phase Transition

Literature states that a phase transition for a network/lattice occurs at $r = 0.5$ [8] [9], where if $r < 0.5$ the system approaches clustering behaviors and if $r > 0.5$ the system approaches consensus among the agents. The simulations tends to agree with this assessment for a uniform distribution within the starting lattice.

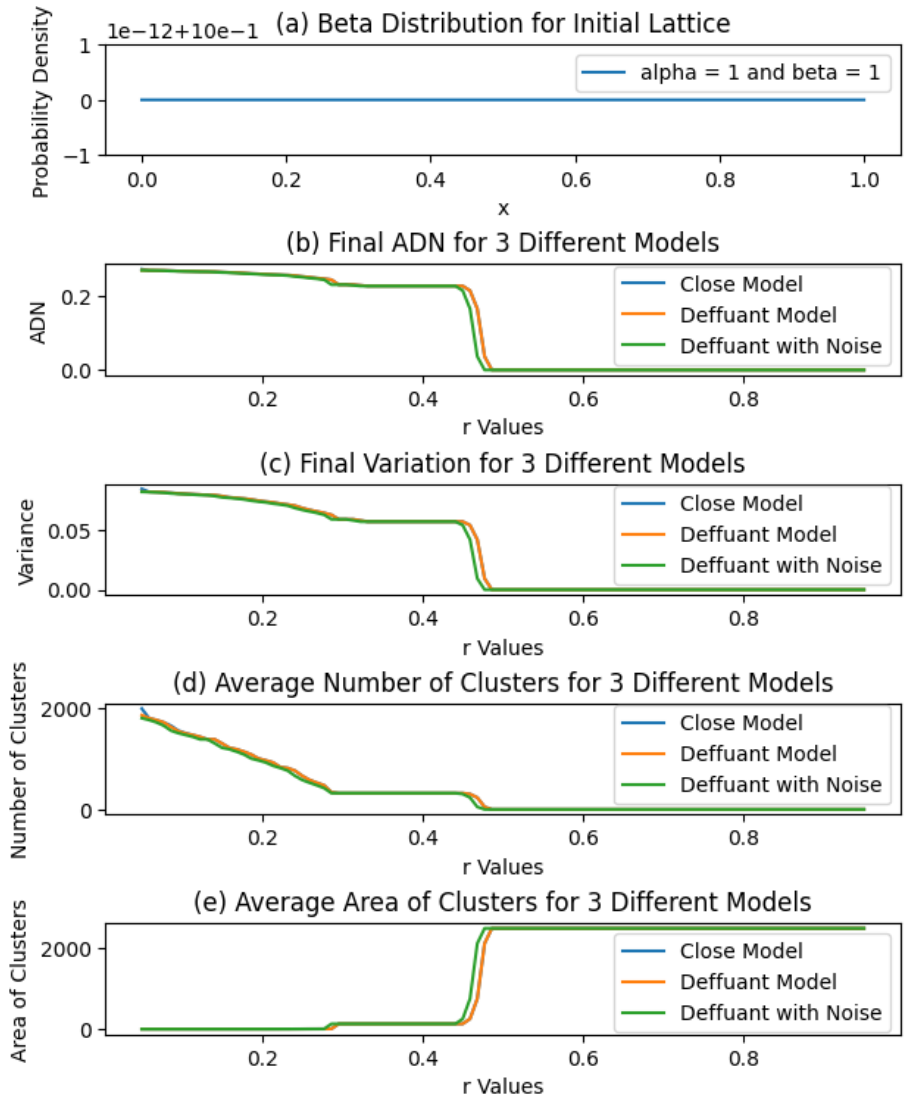


Figure 3.8: Various measurements on a 50 by 50 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 1$ and $\beta = 1$ results in a uniform distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

From the above the phase transition appears to occur at approximately $r = 0.5$ for the close Deffuant model and the normal Deffuant model, however the phase transition occurs slightly earlier for the noisy Deffuant model. This also occurs for other initial lattices that use a beta distribution for generation that is symmetrical around 0.5, as shown below.

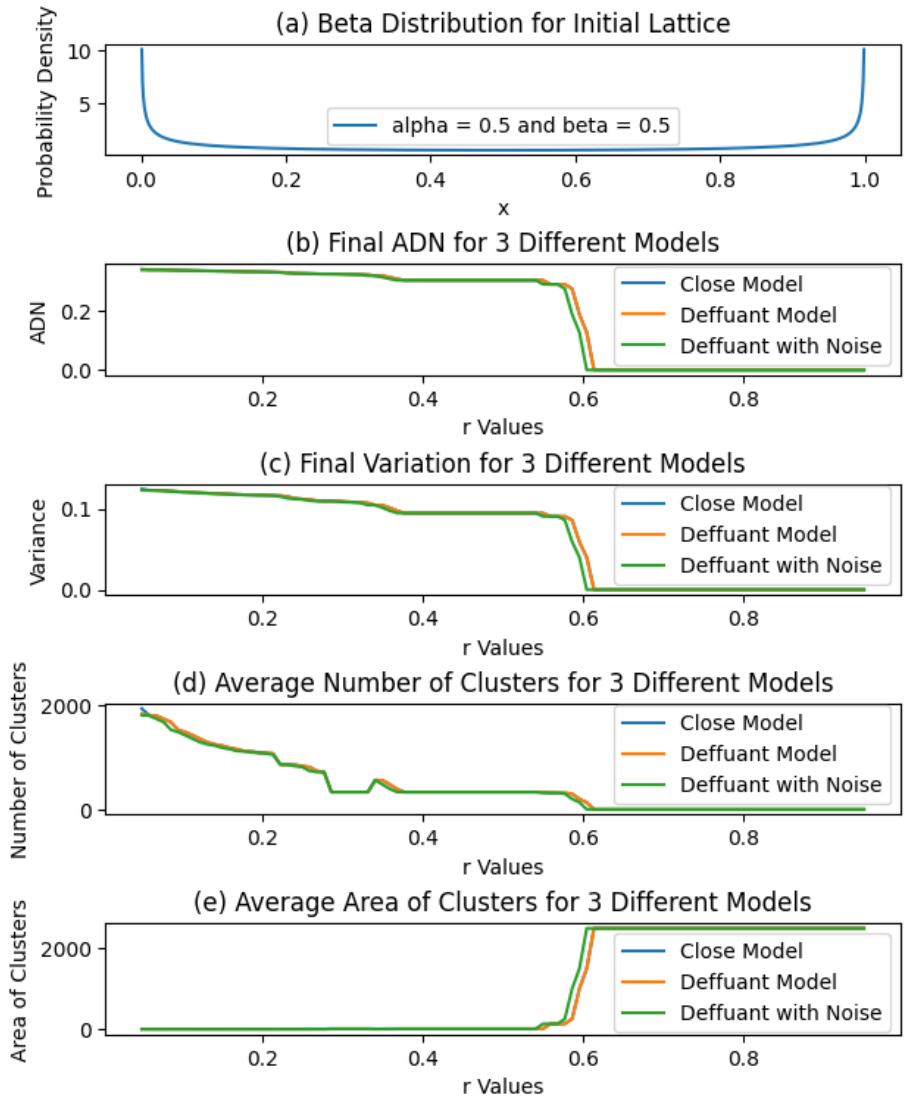


Figure 3.9: Various measurements on a 50 by 50 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 0.5$ and $\beta = 0.5$ results in a more polarised distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

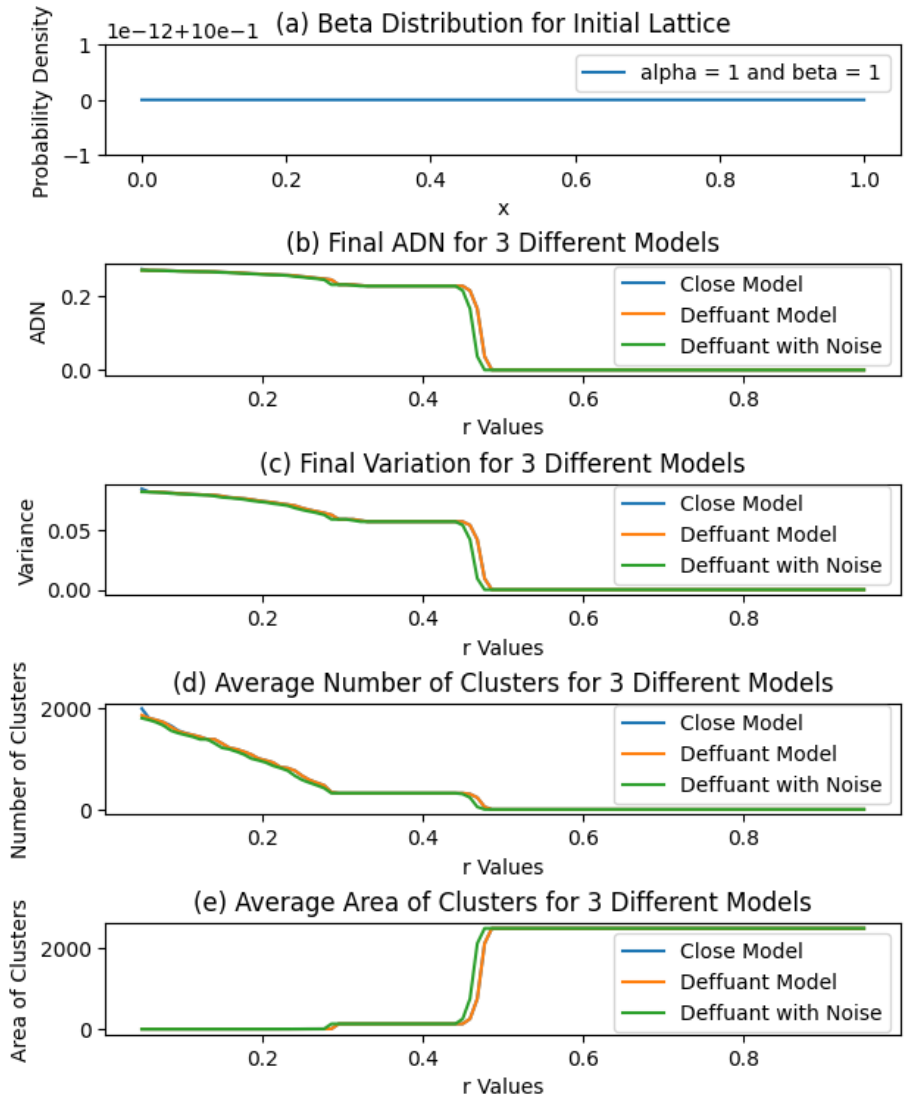


Figure 3.10: Various measurements on a 50 by 50 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 1$ and $\beta = 1$ results in a uniform distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

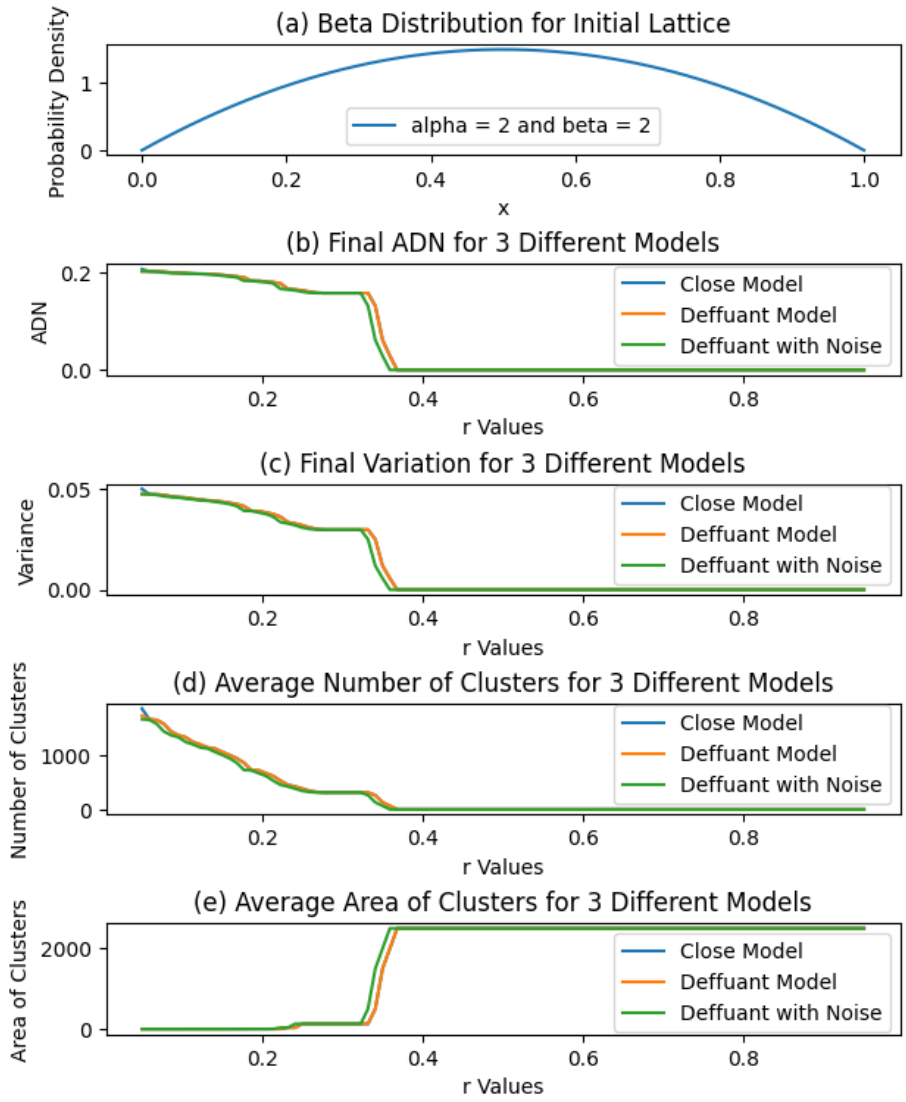


Figure 3.11: Various measurements on a 50 by 50 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 2$ and $\beta = 2$ results in a more centralised distribution for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

While this behaviour is very clear in the case where the beta distribution used to generate the initial lattice is symmetric around 0.5, it is less clear with an asymmetric beta distribution, although it still appears to exhibit the same behaviour.

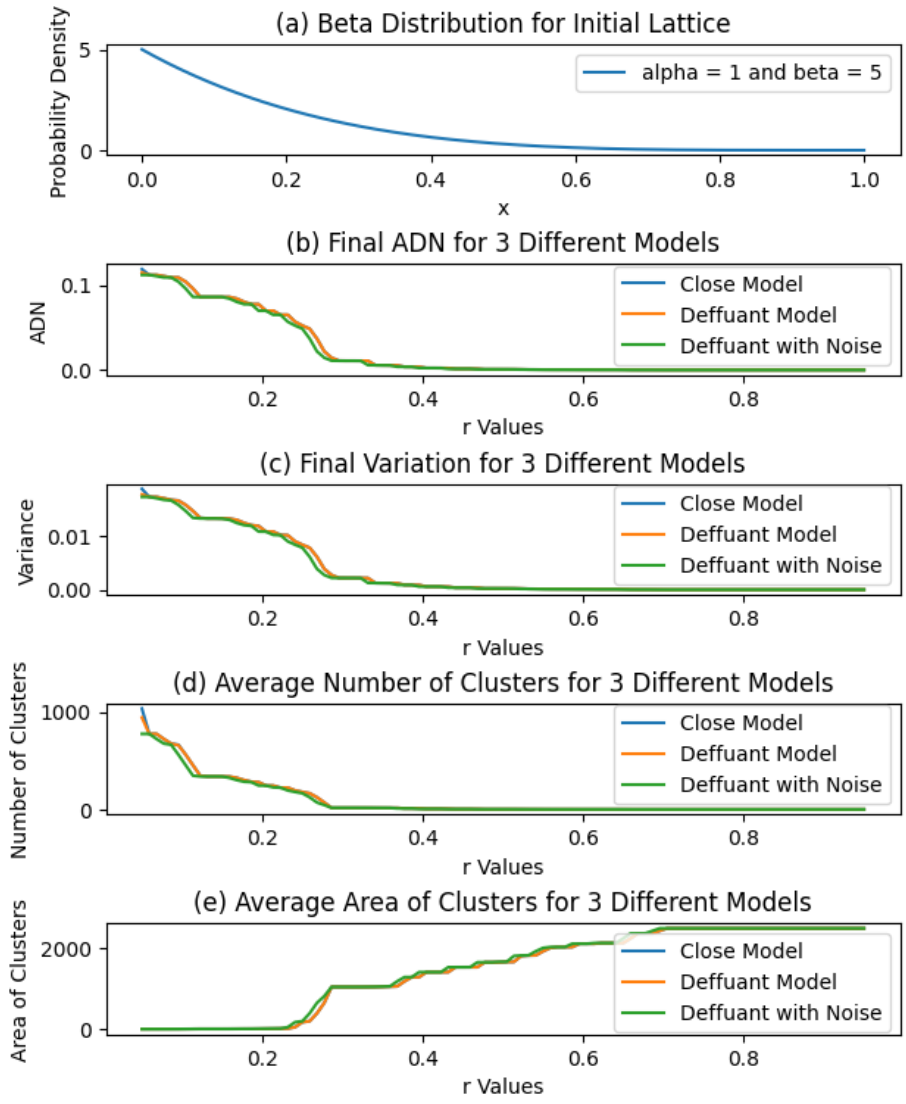


Figure 3.12: Various measurements on a 50 by 50 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 1$ and $\beta = 5$ results in a distribution that skwes towards 0 for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

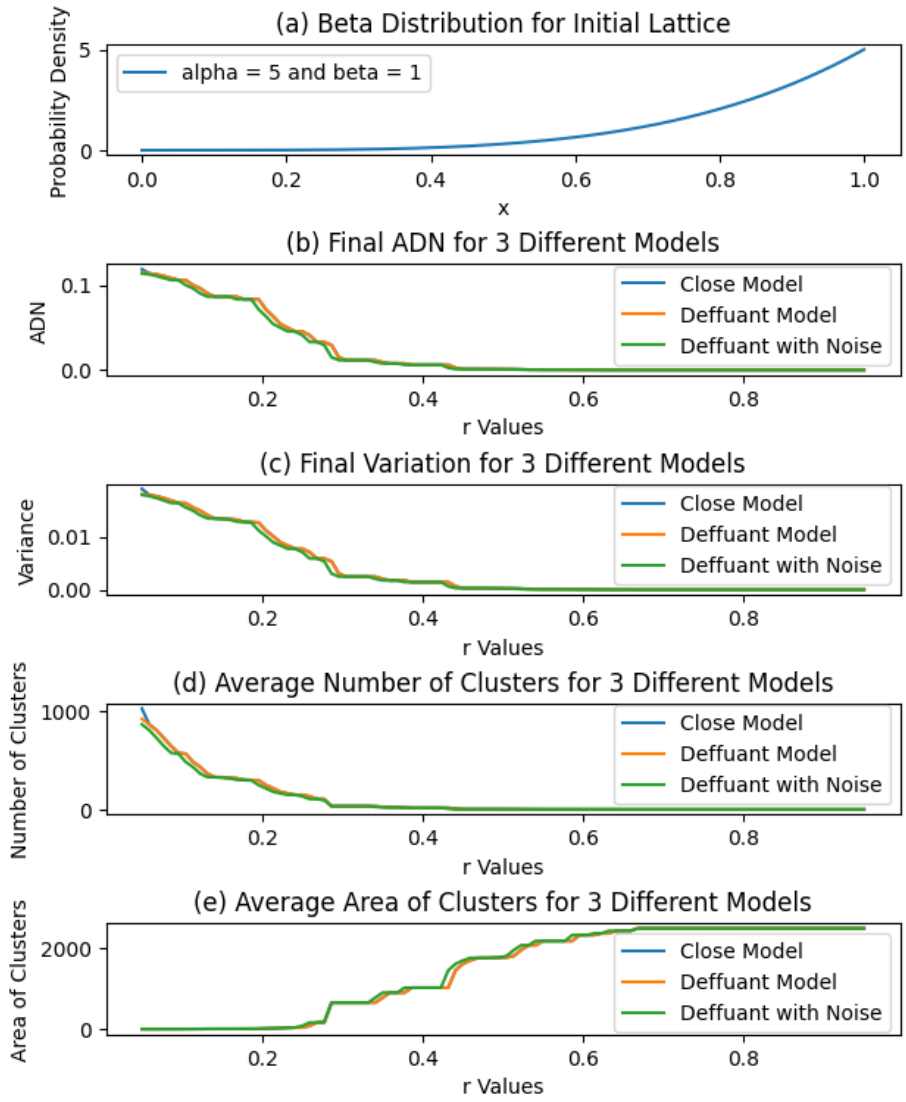


Figure 3.13: Various measurements on a 50 by 50 lattice. The lattice was run through the iteration process 20 separate times (or 20 different simulations as they will be referred to) and then measurements were averaged over those simulations. $\mu = 0.4$ was used for iteration. (a) The beta distribution used to generate the lattice. Using $\alpha = 5$ and $\beta = 1$ results in a distribution that skews towards 1 for the generation of the lattice. (b) The averaged ADN for the three models at the last iteration over different r values. (c) The average variation for the three models at the last iteration over different r values. (d) The average number of clusters for the three models at the last iteration over different r values. (e) The average area of the clusters for the three models at the last iteration over different r values.

This behaviour occurs due to how the system typically reaches equilibrium. For low r values (r values below the phase transition point), the system will organise itself such that sites only take a finite number of values. Take the example from section 2.4, where a value of $r = 0.25$ results in the lattice sites taking one of three distinct values. An example run using the normal Deffuant model is shown below of this situation.

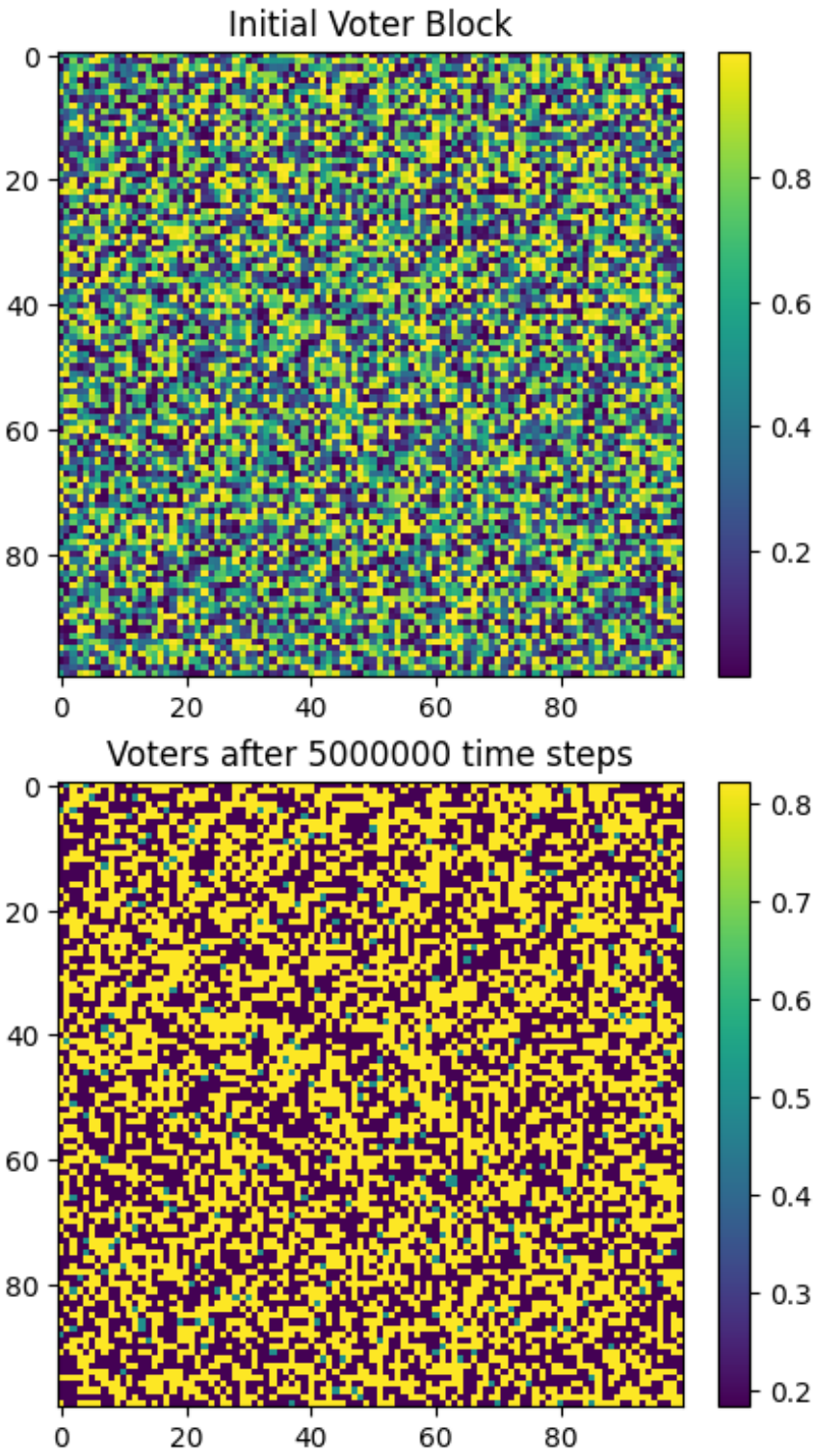


Figure 3.14: The sites in the lattice have one of 3 values, 0.184, 0.503 and 0.821. Initial lattice has $\alpha = 0.5$ and $\beta = 0.5$ for beta distribution generation, dimensions of 100x100 and the model used is the normal Deffuant model with a μ value of $\mu = 0.4$.

From the above, one can see that the magnitude of the difference between 0.184 and 0.503 is greater than 0.25 and likewise for 0.503 and 0.184 and 0.821. So while the system is in the process of reaching equilibrium, values within 0.25 of 0.184, 0.503 or 0.821 will approach those respective values, but cannot reach other values via communication. This is all because $r = 0.25$ in this case. Once r is sufficiently large however, all sites are able to communicate with each other, they may need to communicate with other sites closer in value first but they eventually are able to be influenced by any and all sites and due to the updating rule they will eventually converge to consensus. The value of r this typically happens at is $r = 0.5$ (for uniform starting lattice), as enough sites are able to communicate for consensus. However, with the noisy Deffuant model, r is not limited to being a single value, it has a chance to either be slightly less than or (more importantly for the phase transition)

greater than a mean value. Typically the variation from the mean was made small, so the noisy Deffuant model mostly followed the other two models for most r values. Near the phase transition however, the spread allows r to take values that go over the typical phase transition value. If this occurs enough, the system behaves as if it is over the phase transition value of r, and so the system transitions to consensus earlier than the other two models. The reason it occurs earlier is because consensus cannot be undone between sites due to how the update rule always brings the value of the two sites closer, the update rule cannot take two sites' values further apart.

One may have also noticed that the phase transition is affected by the initial beta distribution used to generate the lattice. If the distribution used is more central, the phase transition occurs for a smaller r value and if the distribution is more extreme on opposite ends (which will be referred to as the extreme case) then the phase transition occurs for a higher r value, as seen in Figure 3.9. For the extreme case, the lattice sites will remain extreme as there are little central sites within range r to help bridge the gap between the extremes. This results in the r value for the phase transition increasing as it needs to be sufficiently large to bridge the two extremes directly, rather than having central sites to "connect" the two extremes. A more central distribution for the initial lattice results in a phase transition at a lower r value as the larger central population brings the more extreme lattice sites closer to central values (assuming the range r at which is large enough to capture some of the extreme sites), and as the central sites greatly outnumber the extreme sites, the central sites end up dominating. This causes consensus for lower r values.

3.4 Conclusion

This implementation of the Deffuant model to a square/rectangular periodic lattice supports the findings of the literatures' conclusions. The phase transition r value is about 0.5 for a uniform lattice using the Deffuant model, as well as for the close Deffuant model. The noisy Deffuant model consistently has a phase transition point lower than that of the other two models. μ values and the size of the lattice (assuming large enough) had little effect on the long term behaviour of the phase transition, instead affecting the number of iterations needed to reach an equilibrium. Above the phase transition, the long term behaviour of the lattice is for it to reach consensus. Below the phase transition, the lattice tended to have lattice sites go to a finite number of values, dependent on the r value. For symmetric but non-uniform probability of values in the starting lattice, the phase transition would either be less than 0.5 (for more centralised distribution) or greater than 0.5 (for more polarised distribution) in the long term. Asymmetric distributions do not seem to exhibit a phase transition like the symmetric distributions do.

Appendix A

Extensions to Deffuant Model

There are various extensions to this model presented in this model that will not be covered but are worth mentioning. One such possible extension is known as the heterogeneous Deffuant–Weisbuch model [2]. This model does not make the assumption that the r value is constant (or the mean is constant in the case of the noisy Deffuant model), instead each r value is unique between agents (hence the term heterogeneous). Traits such as convergence and a bounded mean-square error are seen in this version of the Deffuant model. Similar could also be done with the value of μ, but seeing as the value of μ had little effect on the long term behaviour of the lattice in this report one would expect little to change between the findings of this report and this extension (although this is not confirmed).

Another extension is to assume that agents who do not communicate (that is, sites that are outside the specified r value) instead repel each other away from each other, rather than doing nothing [9]. This is referred to as repulsion and is a good fit for the Deffuant model as it only really had attraction before. Mixes of repulsion and attraction are also an interesting extension of the Deffuant model.

Another possible extension is the implementation of zealots. These are lattice sites that would not change value regardless of communication with other sites. One would expect clusters to form around these sites and has some interesting consequences for other models and different dimensions, such as breaking conservation of magnetisation for dimension less than or equal to 2 [1].

The models in this report can be used on other types of graphs and are not limited to square/rectangular, periodic lattices. Any sort of network or odd-shaped lattice could also be studied. Generation of random graphs with random weightings in between nodes may provide a more realistic scenario than a square lattice.

There are many other extensions to the Deffuant model out there, from hybrid models that combine existing models together, to new takes on old models. These are just some modifications that have been investigated or considered.

Bibliography

[1] C. Castellano, S. Fortunato, and V. Loreto. Statistical physics of social dynamics. *Reviews of Modern Physics*, 81(2):591–646, 2009.

[2] G. Chen, W. Su, W. Mei, and F. Bullo. Convergence properties of the heterogeneous deffuant–weisbuch model. *Automatica*, 114:108825, 2020.

[3] J. Choi, B. Min, and T. Galla. Analysis of a voter model with an evolving number of opinion states. *Physical Review E*, 111(6), 2025.

[4] E. J. de Area Leão Pereira, M. F. da Silva, and H. Pereira. Econophysics: Past and present. *Physica A: Statistical Mechanics and its Applications*, 473:251–261, 2017.

[5] G. Deffuant, D. Neau, F. Amblard, and G. Weisbuch. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 03(01n04):87–98, 2000.

[6] R. A. Donnelly and F. Abdel-Raouf. *Statistics*. Alpha, a member of Penguin Random House LLC, 3rd edition, 2016.

[7] S. Ferrari and F. Cribari-Neto. Beta regression for modelling rates and proportions. *Journal of Applied Statistics*, 31(7):799–815, 2004.

[8] S. FORTUNATO. Universality of the threshold for complete consensus for the opinion dynamics of deffuant¡i¿et al.¡/i¿. *International Journal of Modern Physics C*, 15(09):1301–1307, Nov. 2004.

[9] N. Lanchier and M. Mercer. Deffuant opinion dynamics with attraction and repulsion, 2023.